UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

DEPARTMENT OF COMPUTING AND TECHNOLOGY

EASTER - 2026

PROGRAM: MSC DATA SCIENCE & MSC COMPUTER SCIENCE

ASSIGNMENT 2 (GROUP)

SEMESTER: EASTER 2026

Assignment Objective

To collect, analyze, and evaluate Uganda's historical, observational, and gridded weather datasets aligned with UNMA station locations in order to identify key weather patterns and assess data suitability for predictive modelling in Uganda.

This assignment assesses students' ability to source, preprocess, analyze, and model real-world weather data using publicly available climate datasets, with emphasis on data suitability, feature engineering, and predictive modelling.

Background

The **Uganda National Meteorological Authority (UNMA)** is the national institution mandated to observe, monitor, analyze, and disseminate weather and climate information in Uganda. UNMA operates a network of weather stations across the country, including synoptic, climatological, agro-meteorological, and rainfall stations. Due to limited spatial coverage and restricted public access to raw station data, researchers often complement UNMA observations with satellite and reanalysis datasets such as CHIRPS, ERA5, etc.

Identifying UNMA Weather Stations:
Using UNMA observational datasets is not required for this assignment. Students are expected to use Google Search to identify weather stations operated by UNMA. Appropriate search queries may include terms such as "UNMA weather stations in Uganda", or "Uganda National Meteorological Authority station locations", or "UNMA synoptic stations Uganda". The identified stations will form the spatial reference points for subsequent assignment data analysis and modelling tasks.

Data Requirements

Each group must:

1. Identify at least five (5) UNMA weather stations in Uganda
2. Obtain approximate latitude and longitude for each station
3. Collect at least **two (2)** publicly available gridded weather datasets, each with a minimum temporal coverage of **five (5) consecutive years**, such as CHIRPS (rainfall), ERA5 Reanalysis, Copernicus Climate Data Store, or Open-Meteo historical data. The selected gridded datasets must be spatially and temporally aligned with the chosen UNMA station locations.

Assignment Tasks (25 Marks Total)

1. UNMA Station Identification and Mapping [4 Marks]

a) Identify at least three UNMA weather stations in Uganda

b) Obtain latitude and longitude for each station

c) Create a table summarizing station information

d) Map the station locations using Python and clearly label each station

2. Weather Data Collection [4 Marks]

a) Download at least two gridded weather datasets covering Uganda

b) Extract weather variables corresponding to the selected UNMA station locations

c) Align datasets temporally (daily or monthly)

d) Briefly explain how gridded data approximates station observations

3. Data Preprocessing and Quality Assessment [5 Marks]

a) Handle missing values and inconsistent timestamps

b) Detect and treat outliers and physically unrealistic values

c) Ensure consistency between datasets

d) Discuss data gaps and limitations common in data-scarce regions

e) Save the transformed datasets from above as your Group Name (e.g. Dataset1_Group(a).csv, Dataset2_Group(a).csv, Dataset3_Group(a).csv, etc.)

4. Exploratory Analysis of Ugandan Weather Patterns [5 Marks]

MAM denotes the March–April–May (long rains) season, while SON denotes the September–October–November (short rains) season

a) Using the transformed dataset from question 3 saved as (e.g. Dataset1_Group(a).csv, Dataset2_Group(a).csv, Dataset3_Group(a).csv, etc.), analyze seasonal patterns (e.g. MAM and SON rainfall seasons)

b) Identify interannual variability and extreme events

c) Compare patterns across datasets and stations

d) Use appropriate visualizations to support findings

5. Feature Engineering and Dataset Suitability Analysis [4 Marks]

a) Engineer relevant features for weather modelling (lagged values, rolling means, seasonal indicators)

b) Assess dataset suitability for Ugandan weather prediction

c) Justify which dataset(s) are most appropriate for modelling in Uganda

6. Weather Prediction Model Development [3 Marks]

a) Select one weather variable to predict (e.g. rainfall or temperature)

b) Split data into training and testing sets

c) Train and evaluate a suitable machine learning model

7. Model Saving [0.5 Marks]

a) Save the trained model as a joblib or pickle file

8. Prediction Generation and Export [0.5 Marks]

a) Generate predictions on unseen data

b) Save predictions as a CSV file (e.g. `uganda_weather_predictions.csv`)

Submission Instructions

1. This is a **GROUP ASSIGNMENT** (3–5 students per group). This assignment contributes 25% to the total course grade.
2. Submit one ZIP folder per group containing, i.e.,
   i. Python notebooks (.ipynb)
   ii. Raw datasets collected
   iii. Processed datasets
   iv. Saved model files

3. All notebooks must clearly include group member names and registration numbers
4. **Submission Deadline: 17$^{\text{th}}$ February 2026**

◄ END ►